

NLP Project - Language Specific Finetuning Effects

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[Code on GitHub](#)

Abstract

We propose a framework for investigating cross-lingual transfer and forgetting in Named Entity Recognition (NER) with respect to language similarity in the setting of finetuning multilingual masked language models to specific languages using the LoRA (Hu et al., 2021) approach. As a baseline, we finetune XLM-RoBERTa (Conneau et al., 2020) to the NER objective with equal amounts of labeled training data from 12 languages in our data. In our experiments, we continue finetuning this baseline to 5 different target languages (separately) while monitoring performance evolutions on all the languages. We find that more similar target and test language pairs likely tend to improve their performance when finetuned on target language and evaluated on test language, while dissimilar pairs' performance suffers. However, the performance changes are small, the trends are fairly weak and depend on a few outliers and the specific finetuning and evaluation methods.

1 Introduction

Since the late 90s, significant improvements have been reached in Named Entity Recognition (NER), facilitating the processing of unstructured data. Multilingual model performance is often skewed towards high-resource languages, resulting in finetuning projects that aim to close this gap. Recent research explores the usage of few-shot finetuning (Lauscher et al., 2020), vocabulary re-building (Arhipov et al., 2019), and retraining BERT models from scratch (Yohannes and Amagasa, 2022) on low-resource languages. They demonstrate that adopting language models to low resource settings can improve performance (Eisenschlos et al., 2019). Less attention seems to be directed at exploring how such finetuning affect performance on non-target domains, i.e. when forgetting occurs in relation to how "far" the domain is from the target

domain. In the original Universal NER paper (Mayhew et al., 2024), XLM-RoBERTa is finetuned on all the dataset languages and subsequently tested on all these languages, resulting in a matrix of F1-scores. The authors' results suggest that finetuning cannot leverage cross-lingual transfer for very dissimilar languages. As an example, they observe that a model finetuned on Maghrebi-Arabic-French performs poorly on European languages, while a model finetuned on a European language achieves $F1 \sim .6$ on other European languages, observing some amount of cross-lingual transfer. It is worth noting that these are all 0-shot scenarios, i.e. the model was never presented with labeled NER data languages other than the language it was finetuned with.

A scenario that allows us to observe forgetting as well as cross-lingual transfer simultaneously is one in which a "generalist" model (i.e. that has seen labeled NER data from all target languages equally), is further finetuned to one of these languages. Indeed, past research has shown that domain-specific finetuning and continued pretraining of a generalist model can lead to a drop in performance on out-of-domain data (Dorfner et al., 2024). This poses the question: *As a multi-domain model is finetuned more and more to one domain, how much does performance degrade in the off-domain? Can the "similarity" of the domains inform us how much we can expect performance to drop?* We believe that exploring these questions is meaningful because the findings enable a more conscious decision when finetuning models, being aware of the tradeoff one is making. Thus, we pursue the research question: **"How does language specific fine tuning of XLM-RoBERTa affect performance in the downstream task of NER with respect to language similarity in terms of (a) forgetting (b) cross lingual transfer?"** Therefore, we define a baseline as a "generalist NER model", finetuned with all languages. Then we continue finetuning

the baseline to the 5 languages in separate experiments, observing the evolution of performance on 12 languages (Slovak, English, Swedish, Norwegian (Nynorsk), Hebrew, Romanian, Portuguese, German, Mandarin, Croatian, Serbian, Danish).

Our hypothesis is that the positive effects of cross-language transfer will outweigh the consequences of forgetting in earlier epochs, especially for languages similar to the target language, while the trend will reverse in later epochs, when the model should be mainly specialized on the target language.

2 Data

The annotated and BIO-encoded NER data for each of the 12 languages in the scope of the paper, listed in [section 1](#), is sourced from the UniversalNER dataset ([Mayhew et al., 2024](#)). The data mostly follows the 3-entity scheme PERSON, ORGANIZATION, LOCATION, however, cases of entities classified as OTHER occur in Croatian and Serbian. Such entities are discarded, and the corresponding tokens are assigned the O label (not-an-entity).

For the majority of languages, train, development, and test files are available, with a few exceptions presenting just one data file.

Exactly one train and one test sets are derived for each language: the development set, when present, is merged with the train set. In the case of single-file languages data, we split it into two partitions. For target language data, 80% of the sentences are allocated to the train set, as this constitutes future finetuning data, with the remaining 20% constituting the test set; for all other languages, not required for language-specific model tuning, it is possible to assign more data (50%) to the test set.

The amount of labeled sentences vastly varies across datasets, and is therefore standardized to ensure a fair performance comparison across languages. When loading the data, the sizes of the smallest datasets are monitored, stored, and used to perform stratified resampling of the datasets to ensure that the same amount of sentences is present in all test sets (491), all target language train sets (4497), and all non-target language train sets (500), while approximately retaining the original Entity Density (ED).

3 Defining Language Distance Metric

To investigate whether cross-lingual transfer and forgetting are related to linguistic distance, we

LANG	TRAIN				TEST	
	Orig. ED	$k = 10$	$k = 500$	$k = 4497$	Orig. ED	$k = 491$
chinese	0.1203	0.0876 (-0.0327)	0.1271 (+0.0068)	- *	0.1176	0.1183 (+0.0007)
danish	0.0548	0.0585 (+0.0037)	0.0580 (+0.0032)	0.0548 (+0)	0.0643	0.0646 (+0.0003)
english	0.0516	0.0407 (-0.0109)	0.0510 (-0.0006)	0.0542 (+0.0026)	0.0669	0.0678 (+0.0009)
romanian	0.1044	0.0704 (-0.0340)	0.1119 (+0.0072)	0.1051 (+0.0007)	0.0983	0.0953 (-0.0030)
slovak	0.0439	0.0482 (+0.0043)	0.0459 (+0.0020)	0.0420 (-0.0019)	0.1089	0.1053 (-0.0036)

*no sampling performed, all datapoints retained.

Table 1: ED for original dataset and for a sample of k sentences (with change from original) - Target Languages

LANG	TRAIN				TEST	
	Orig. ED	$k = 10$	$k = 500$		Orig. ED	$k = 491$
croatian	0.0806	0.0823 (+0.0017)	0.0858 (+0.0052)		0.0856	0.0870 (+0.0014)
german	0.0712	0.0847 (+0.0135)	- *		0.0689	0.0690 (+0.0001)
hebrew	0.0794	0.1010 (+0.0216)	0.0717 (-0.0077)		0.0721	- *
norwegian	0.0579	0.0448 (-0.0131)	0.0539 (-0.0040)		0.0535	0.0581 (+0.0046)
portuguese	0.0794	0.1173 (+0.0379)	0.0911 (+0.0117)		0.0777	0.0939 (+0.0162)
serbian	0.0984	0.0857 (-0.0127)	0.1007 (+0.0023)		0.1071	0.1068 (-0.0003)
swedish	0.0263	0.0261 (-0.0002)	0.0281 (+0.0018)		0.0325	0.0328 (+0.0003)

*no sampling performed, all datapoints retained.

Table 2: ED for original dataset and for a sample of k sentences (with change from original) - non-Target Languages

construct a language-distance dictionary using the eLinguistics online language comparison tool, as an external measurement. The score is a numerical distance between two languages, where a lower value means higher linguistic proximity ([eLinguistics.net, 2026](#)).

We chose this method because it provides a consistent, ready-to-use numerical distance score for all language pairs in our experiment. Other, more commonly adopted language similarity scores were considered, but later discarded. The main candidate, URIEL+ ([Khan et al., 2025](#)), presents numerous missing entries in the syntax feature vector of multiple languages, with only English having all values specified, compared to only 28% for Serbian, less than 12% for Croatian, and Slovak not having any.

The resulting distances are interpreted as approximate indicators of lexical-genetic relatedness rather than complete measures of language similarity, since they do not fully capture syntax or morphology. Further details on the calculation method are provided in [Appendix C](#).

4 Finetuning Methodology

All our experiments are conducted using XLM-RoBERTa ([Conneau et al., 2020](#)), a widely used model for the NER task. Instead of tuning all weights in a full finetune, we train LoRA adapters ([Hu et al., 2021](#)), hypothesizing that the changes needed to adapt the transformer’s weight matrices to a new language live in a space of lower rank. See [Appendix D](#) for a more in-depth discussion of the advantages and technicalities of LoRA.

4.1 Baseline

As a baseline we create a "generalist NER model", which has been finetuned with labeled NER training data from all languages in the data (500 samples each). We train a LoRA adapter with the following hyperparameters, with Learning Rate and Batch Size taken from the course material, and the remaining being sensibly adapted to the use-case.

Hyperparameter	Value
LoRA Rank	32
LoRA α	32
Learning Rate	0.0001
LR Scheduler Type	constant
Epochs	3
Batch Size	64

Table 3: Baseline Hyperparameters

4.2 Language-Specific Finetunes

For each of the 5 target languages, we continue the training of the baseline model's LoRA adapter to obtain the language-specific LoRA adapter. The training data includes 4497 examples from the respective target language.

Hyperparameter	Value
LoRA Rank	32
LoRA α	32
Learning Rate	0.0001
LR Scheduler Type	linear
Epochs	15
Batch Size	64

Table 4: Language-Specific Hyperparameters

5 Evaluation Methodology

The evaluation process involves language-specific metric computation for each individual fine-tuning epoch of the given model. This aims to provide a more in-depth report, monitoring the evolution of the model's performance as the fine-tuning progresses. Span-F1 score serves as the main metric for both decision-making and result presentation, and, while still extracted, Precision, Recall, and Accuracy take on a background role. This results in 20 sets of metric values for each of the 12 languages, and is repeated for all 5 language-specific models, as well as for the baseline model.

The results presented in [section 6](#) are derived from such metric values through different approaches, namely:

- Language-specific performance comparison between fine-tuned model and baseline model to quantify effects of language-specific post-training on both the model's target-language, and non-target-languages.
- Given a model, the Spearman Rank Correlation between each language's performance metrics and its similarity to the model's target-language to observe the relationship between the two measures.
- Analyzing performance changes through epochs to determine if and when during post-training a model "overly-specializes" on its target-language, significantly worsening performance even on closely related languages.

6 Results

Target	eng	slk	dan	rom	chi	swe	nor	heb	por	ger	hrv	srb
all	.65	.80	.82	.70	.68	.84	.85	.51	.82	.79	.87	.87
eng	.74	.78	.82	.58	.68	.87	.84	.55	.82	.78	.90	.89
slk	.65	.87	.82	.76	.63	.85	.86	.58	.82	.75	.89	.90
dan	.72	.80	.86	.62	.68	.86	.87	.60	.85	.79	.92	.91
rom	.58	.79	.80	.85	.63	.85	.84	.50	.80	.76	.88	.87
chi	.67	.80	.83	.59	.85	.86	.86	.62	.84	.81	.88	.88

Table 5: Span-F1-scores by target language (rows) and test language (columns). Note that "all" refers to the baseline model described in [subsection 4.1](#). **Highlighted** are F1-scores for which the target language and the test language are the same.

As visible in [Table 5](#), the performance for a given test set (column) is highest for the model (row) where that language was also the training language (**highlighted**). This indicates that further finetuning of the baseline does, in fact, improve performance for the target across the board, from a moderate .04 F1 increase for Danish to a notable .17 F1 increase for Chinese.

Looking at [Figure 2](#), one can observe that the most substantial performance differences occur during the first few epochs, after which scores stabilize. Therefore, we surmise that fewer epochs are needed to reach comparable results to the ones shown, saving compute.

Some cases of non-target language benefiting from specialized finetuning in another language can be observed. Focusing on Chinese finetuning, the process considerably improves the F1 score for

Chinese, as well as for Hebrew, more surprisingly. The Danish-specific model exceeds baseline scores for the majority of languages, with greater changes for Hebrew, English, Croatian, and Serbian.

Generally, the chosen workflow seems to not cause any catastrophic forgetting in non-target languages, commonly culminating in stable, or lightly improved performance. Some exceptions do occur, mainly involving Romanian, which suffers a -0.11 F1 shift for Chinese finetuning, and a lighter performance decrease on other language-specific models, with the exception of the Slovak-specialized model, where it improves almost as much as Slovak itself. However, it should be noted that training in Romanian does not improve Slovak performance. Logically, this can be due to the difference in baseline performance for the two languages: the finetuning is more likely to elevate the more modest 0.70 F1 recorded for Romanian, than the 0.80 F1 for Slovak, which determines a narrower margin for improvement.

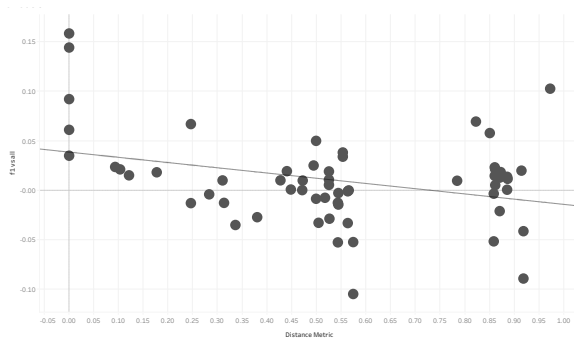


Figure 1: Scatterplot of Language Distance Metric (x) and Δ F1-score (y) (the difference in performance on a given language between the model trained on a set target language and the baseline model, as seen in Figure 2). Each point represents a combination of target language and test language.

The downwards trend¹ in Figure 1, paired with a light yet present Spearman Rank Correlation of -0.278 , seems to suggest that, indeed, performance tends to worsen the more language distance grows. It is worth noting that all datapoints with Distance Metric ≥ 0.6 involve either Chinese or Hebrew, the two languages most dissimilar to others. Interestingly, when removing the 5 datapoints with Distance Metric = 0.0 (instances where the same language is used for training and testing), the trend line flattens, and the coefficient p-value becomes statistically insignificant. This is also the

case when removing exclusively the two datapoints corresponding to Chinese-Chinese and Romanian-Romanian train-test language pairs, in line with the observation that Chinese and Romanian seem to benefit the most from language-specific finetuning compared to other target languages, considerably swerving the resulting regression line.

7 Conclusions and Further Research

While some evidence of correlation between the language similarity of target and test language and the performance change between target-language finetuned model and baseline is found, further research is needed to discern a clearer pattern concerning forgetting and cross-lingual transfer in relation to language similarity. As discussed in section 6, the significance of this finding vanishes when removing only two datapoints. We therefore end our analysis on an ambiguous note. Other finetuning methods should be explored to clearly separate the effects of post-training from the underlying influence of language similarity on model forgetting and cross-lingual transfer. We believe that the chosen LoRA approach does result in a model limited in its capacity to learn from recent epoch while potentially forgetting information related to older epochs, due to the restricted trainable parameter space.

Finally, a language similarity score that is consistent, considers multiple language features, and is well populated for all languages, is needed to be able to fully investigate the interactions of similar and dissimilar languages with Natural Language Processing tasks. Therefore, more research has to be done in this area, in order to be able to reach further conclusions.

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¹Linear Fit: coefficient = -0.061 , pval = 7.3×10^{-3}

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A Individual Member Contributions

All members discussed and decided all main considerations together. Code was written, reviewed and rewritten by all members at different times and at different points. Visualizations were made with decisions and input by all members. Report was written and reviewed by all members with specific sections being the result of numerous changes committed by different members at different points.

B Chatbot usage

Chatbot assistance was involved in the following tasks:

- Obtaining documentation for Hugging Face methods, including hyperparameter description.
- Formatting and creation of $\LaTeX$ elements, such as tables.
- Finding alternative words.
- Ensuring proper code for loading and usage of LoRA weights (avoiding stacking LoRA adapters during evaluation).
- Finding relevant research papers.
- Help with plotting in Tableau.
- Assistance in creating a README instructions file for GitHub.
- Creation of the "Nearly Lucid Penguin" [Figure ??](#).

C Language Similarity Calculation

We collect language-distance scores from the eLinguistics online language comparison tool. The tool estimates linguistic distance using a lexical comparison method based on a small set of 18 basic vocabulary items, which are treated as linguistic "genetic markers" (Beaufils, 2015). These words are selected because they are expected to be historically stable, resistant to borrowing, resistant to semantic shifts, and relatively resistant to phonetic erosion.

The calculator converts the selected words into sound classes and compares them using preestablished sound-correspondence probability scores. So, the distance is not based only on surface-level spelling similarity; it attempts to account for systematic sound correspondences between historically related languages. The individual lexical comparisons are then combined into a single distance value between each pair of languages.

For example, English–Danish has a relatively low distance of 24.6, while English–Chinese has a much higher distance of 88.5. Similarly, Slovak is close to Serbian and Croatian, with distances of 9.3 and 12.1 respectively. These examples suggest that the extracted values broadly reflect expected linguistic relationships, such as the similarity between closely related Slavic languages and the larger distance between Indo-European languages and Chinese.

D LoRA

In our finetuning methodology, we employ the LoRA approach as originally proposed in (Hu et al., 2021). What follows is a motivation for using this method, alongside a brief summary of the technicalities and how we chose our parameters.

D.1 Why use LoRA?

One frequent NLP task is the adaptation of a generalist model to a specific task or domain with the aim of significant performance improvements within that domain. Traditional methods such as full finetuning, tune all trainable parameters of the base model. While this may be acceptable for smaller models, it becomes a challenge for bigger models: XLM-RoBERTa occupies ~ 2.2 GB, meaning that 5 specialist models would already consume ~ 11 GB, making model sharing and deployment a challenge in some settings. Prefix Tuning (Li and Liang, 2021) shortens the available context length

and Adapter Finetuning (Houlsby et al., 2019) introduces extra inference latency, rendering these methods suboptimal as well. LoRA circumvents these challenges elegantly by tuning only a small set of trainable parameters² that can be stored and shared independently from the base model. Additionally, these weights can be fused with the the model to eliminate inference latency entirely.

D.2 Inner Workings of LoRA

LoRA rests on the observation that for many finetuning tasks, the updates $\Delta W^{d \times k}$ made to the weight matrices $W^{d \times k}$ are of intrinsically lower rank r than the full matrix. This "low intrinsic rank hypothesis" (Aghajanyan et al., 2021) suggests that the update ΔW can be factorized into two smaller matrices A, B without losing much information.

The Linear Algebra Approach: Singular Value Decomposition (SVD) Such computations are conducted explicitly via SVD in Linear Algebra by removing dimensions with small singular values.

$$W = U \Sigma V^T$$

Here, $U \in \mathbb{R}^{d \times d}$, $V \in \mathbb{R}^{k \times k}$ are square matrices and $\Sigma \in \mathbb{R}^{d \times k}$ is a diagonal matrix with positive singular values, usually arranged descending. One can observe that the SVD for the matrix $\Delta W = W' - W$ representing the difference between the base model weights and the fully-finetuned model's weights typically has a few large singular values $\sigma_1, \sigma_2, \dots, \sigma_r$ meaning that these dimensions influence the final matrix strongly, while most other dimensions $\sigma_{r+1}, \sigma_{r+2}, \dots, \sigma_{\min(d,k)}$ have a minimal effect. Thus, by keeping only the r largest singular values and their corresponding singular vectors in the U, V matrices, truncating all other values and vectors, we can obtain a low-rank factorization of ΔW :

$$\Delta W_r \approx U_r \Sigma_r V_r^T$$

What Happens in Practice In practice, the actual difference in the weight matrices of the finetuned and the base model $\Delta W = W' - W$ is unknown initially, so it is not possible to perform an SVD explicitly on the matrix. Instead, one simply defines

$$\Delta W = A \times B$$

with $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times k}$, hypothesizing that backpropagation will tune the entries of A and B

²(approx. 21 MB large for XLM-RoBERTa with $r = 32$, targetting Query, Key, Value, and Dense layers)

such that they represent a good approximation of the SVD of ΔW .

D.3 LoRA Parameters

- r : represents the rank of the matrix factorization. Suppose $\Delta W \in \mathbb{R}^{3 \times 3}$, $r = 1$, then $A \in \mathbb{R}^{3 \times 1}$, $B \in \mathbb{R}^{1 \times 3}$. Low values e.g. $r = 2$ assume a small intrinsic rank of the matrix ΔW while higher values e.g. $r = 32$ allow for more tunable parameters, coming closer to a full tinetune. In our experiments we find that $r = 32$ yields better performance than $r = 8$ with the size of the LoRA weights measuring only ~ 21 MB per model. Below is a graphical illustration of a $r = 1$ factorization of a 3×3 weight matrix:

$$\Delta W = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix} = \underbrace{\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}}_{A \in \mathbb{R}^{3 \times 1}} \underbrace{\begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}}_{B \in \mathbb{R}^{1 \times 3}}$$

- α : is the LoRA scaling parameter, a scalar with which the part of the activation h corresponding to the LoRA weights BA is multiplied:

$$h = Wx + \Delta Wx = Wx + \alpha(BA)x$$

Note that the PEFT library implementation we use automatically divides by r , i.e. $h = Wx + \frac{\alpha}{r}(BA)x$ which is done to make the magnitude of the weight adjustments agnostic to the chosen rank r because the variance of the product BA has been shown to scale with r . We use $r = 32$ after some initial testing.

- *target_modules*: allows to specify the parts of the transformer to which LoRA parameters should be added. While the original paper (Hu et al., 2021) adds weights only to the Query and Value matrices of the attention process, we also consider the Key matrices as well as the dense MLP layers that lie outside the attention process.

E Extra Figures



		Epoch														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
chi	chi	0.1020	0.1220	0.1420	0.1610	0.1630	0.1590	0.1630	0.1630	0.1660	0.1720	0.1700	0.1740	0.1810	0.1690	0.1710
	dan	0.0100	0.0070	0.0120	0.0240	0.0190	0.0180	0.0130	0.0160	0.0210	0.0160	0.0060	0.0090	0.0090	0.0080	0.0070
	eng	0.0060	0.0140	0.0080	0.0080	0.0090	0.0060	0.0100	0.0080	0.0140	0.0190	0.0180	0.0280	0.0180	0.0200	0.0220
	ger	0.0110	0.0190	0.0180	0.0140	0.0290	0.0130	0.0170	0.0220	0.0200	0.0210	0.0210	0.0230	0.0110	0.0180	0.0220
	heb	0.0810	0.0730	0.1060	0.0930	0.0960	0.1090	0.1010	0.0980	0.1280	0.1110	0.1040	0.1100	0.1160	0.1040	0.1120
	hrv	0.0080	0.0280	0.0320	0.0240	0.0260	0.0350	0.0250	0.0230	0.0290	0.0170	0.0220	0.0250	0.0210	0.0200	0.0160
	nor	-0.0140	-0.0120	0.0200	0.0140	-0.0050	0.0150	0.0120	0.0140	0.0100	0.0000	-0.0050	0.0090	0.0090	0.0050	0.0100
	por	-0.0020	0.0010	0.0130	0.0070	0.0130	0.0100	0.0140	0.0200	0.0210	0.0090	0.0090	0.0140	0.0150	0.0150	0.0180
	rom	0.0130	-0.0480	-0.0460	-0.0840	-0.0920	-0.0950	-0.1070	-0.1090	-0.0910	-0.1060	-0.1210	-0.1150	-0.1150	-0.1090	-0.1080
	slk	-0.0180	0.0030	-0.0010	-0.0010	0.0030	0.0050	0.0020	0.0080	0.0030	-0.0120	-0.0100	-0.0040	-0.0110	-0.0090	-0.0060
	srb	0.0080	0.0190	0.0190	0.0150	0.0190	0.0220	0.0220	0.0160	0.0130	0.0120	0.0110	0.0100	0.0110	0.0130	0.0120
	swe	0.0120	0.0180	0.0350	0.0180	0.0310	0.0320	0.0230	0.0130	0.0230	0.0090	0.0220	0.0270	0.0190	0.0190	0.0190
	chi	-0.0500	-0.0090	-0.0160	-0.0490	-0.0110	-0.0530	-0.0260	-0.0050	-0.0330	-0.0140	-0.0170	-0.0140	-0.0110	-0.0010	-0.0030
	dan	0.0200	0.0300	0.0270	0.0340	0.0300	0.0270	0.0340	0.0380	0.0410	0.0460	0.0400	0.0430	0.0360	0.0400	0.0410
	eng	0.0510	0.0820	0.0780	0.0710	0.0720	0.0590	0.0560	0.0820	0.0590	0.0650	0.0650	0.0540	0.0680	0.0730	0.0710
dan	ger	-0.0040	-0.0190	-0.0240	-0.0150	0.0080	0.0040	0.0010	-0.0030	-0.0090	-0.0010	-0.0010	-0.0050	0.0040	0.0030	0.0030
	heb	0.0380	0.0650	0.0630	0.0360	0.0410	0.0620	0.0720	0.0830	0.0770	0.0830	0.0790	0.0850	0.0850	0.0870	0.0880
	hrv	-0.0010	0.0140	0.0180	0.0120	0.0520	0.0570	0.0510	0.0540	0.0310	0.0390	0.0460	0.0540	0.0510	0.0490	0.0490
	nor	0.0190	0.0010	0.0150	0.0170	0.0070	0.0160	0.0170	0.0040	0.0270	0.0280	0.0280	0.0220	0.0200	0.0260	0.0290
	por	0.0210	0.0260	0.0350	0.0250	0.0210	0.0160	0.0240	0.0340	0.0150	0.0250	0.0190	0.0240	0.0300	0.0330	0.0320
	rom	0.0300	0.0350	0.0050	-0.0540	-0.0700	-0.1050	-0.0520	-0.0450	-0.0630	-0.0600	-0.0960	-0.0860	-0.0810	-0.0710	-0.0720
	slk	-0.0180	-0.0200	-0.0090	-0.0480	-0.0050	-0.0220	-0.0090	-0.0040	-0.0170	-0.0200	-0.0050	-0.0080	-0.0080	-0.0120	-0.0080
	srb	0.0080	0.0160	0.0080	0.0130	0.0400	0.0400	0.0500	0.0500	0.0320	0.0430	0.0410	0.0440	0.0420	0.0430	0.0420
	swe	0.0100	0.0050	0.0240	0.0040	0.0300	0.0440	0.0290	0.0280	0.0240	0.0190	0.0230	0.0170	0.0200	0.0230	0.0200
	chi	-0.0280	-0.0080	0.0140	-0.0100	-0.0020	0.0110	-0.0060	0.0190	0.0170	0.0010	0.0090	0.0000	-0.0050	0.0000	0.0000
	dan	-0.0330	-0.0420	-0.0340	-0.0300	-0.0240	-0.0100	-0.0060	0.0050	-0.0100	0.0020	-0.0060	-0.0020	0.0010	-0.0020	0.0000
	eng	0.0700	0.0630	0.0950	0.0770	0.0810	0.0890	0.0900	0.0910	0.0980	0.0910	0.1210	0.1040	0.1100	0.1060	0.0980
	ger	-0.0310	-0.0310	-0.0170	-0.0370	-0.0130	-0.0130	-0.0010	0.0020	-0.0040	-0.0060	-0.0100	-0.0070	-0.0060	-0.0080	-0.0050
	heb	-0.0430	-0.0410	-0.0030	-0.0090	-0.0030	0.0380	0.0160	0.0570	0.0380	0.0360	0.0500	0.0480	0.0430	0.0330	0.0410
	hrv	-0.0220	0.0130	0.0180	-0.0110	0.0120	0.0180	0.0280	0.0290	0.0280	0.0290	0.0360	0.0310	0.0250	0.0270	0.0280
eng	nor	-0.0570	-0.0780	-0.0590	-0.0730	-0.0460	-0.0170	-0.0200	-0.0260	-0.0370	-0.0530	-0.0190	0.0000	-0.0210	-0.0120	-0.0040
	por	-0.0060	-0.0010	-0.0120	-0.0090	0.0100	-0.0050	-0.0280	0.0100	0.0030	0.0070	0.0060	0.0070	0.0110	0.0060	0.0030
	rom	-0.0110	-0.0610	-0.1020	-0.0760	-0.0990	-0.1120	-0.1470	-0.1140	-0.1330	-0.1030	-0.1260	-0.1250	-0.1180	-0.1240	-0.1160
	slk	-0.0710	-0.0420	-0.0340	-0.0450	-0.0370	-0.0250	-0.0160	-0.0290	-0.0360	-0.0300	-0.0170	-0.0290	-0.0280	-0.0270	-0.0270
	srb	-0.0170	-0.0050	0.0070	-0.0190	0.0100	0.0170	0.0110	0.0250	0.0200	0.0240	0.0230	0.0170	0.0150	0.0190	0.0180
	swe	-0.0190	-0.0180	-0.0080	-0.0170	0.0040	0.0200	0.0090	0.0250	0.0150	0.0140	0.0060	0.0330	0.0280	0.0280	0.0330
	chi	-0.0190	0.0050	-0.0080	-0.0150	-0.0430	-0.1040	-0.0450	-0.0270	-0.0520	-0.0720	-0.0620	-0.0330	-0.0560	-0.0380	-0.0470
	dan	-0.0200	0.0020	-0.0120	0.0080	-0.0100	-0.0110	-0.0050	-0.0240	-0.0110	-0.0170	-0.0180	-0.0200	-0.0180	-0.0120	-0.0180
	eng	-0.0530	-0.0010	-0.0360	-0.0350	-0.0570	-0.0690	-0.0460	-0.0300	-0.0760	-0.0110	-0.0740	-0.0850	-0.0680	-0.0720	-0.0680
	ger	-0.0200	-0.0260	-0.0290	-0.0300	-0.0280	-0.0290	-0.0370	-0.0200	-0.0350	-0.0340	-0.0260	-0.0260	-0.0230	-0.0260	-0.0290
	heb	0.0360	0.0300	0.0100	0.0110	0.0300	0.0200	-0.0080	-0.0030	0.0130	0.0030	0.0050	0.0000	0.0050	0.0000	-0.0040
	hrv	0.0010	-0.0090	0.0030	-0.0150	0.0030	-0.0090	-0.0150	0.0000	0.0060	0.0070	0.0060	0.0130	0.0100	0.0060	0.0080
	nor	-0.0230	-0.0070	-0.0210	-0.0140	0.0020	0.0080	-0.0060	-0.0120	-0.0150	0.0060	-0.0070	-0.0100	-0.0030	-0.0050	-0.0030
	por	-0.0400	-0.0360	-0.0250	-0.0310	-0.0140	-0.0260	-0.0360	-0.0380	-0.0330	-0.0280	-0.0140	-0.0270	-0.0150	-0.0240	-0.0170
	rom	0.1060	0.1340	0.1210	0.1370	0.1580	0.1530	0.1480	0.1410	0.1540	0.1470	0.1510	0.1570	0.1530	0.1530	0.1520
rom	slk	-0.0020	0.0170	-0.0020	0.0040	0.0070	-0.0200	-0.0290	-0.0120	-0.0100	-0.0150	-0.0180	-0.0110	-0.0100	-0.0110	-0.0130
	srb	-0.0060	-0.0060	-0.0090	-0.0190	0.0100	0.0060	-0.0180	0.0020	-0.0040	0.0090	0.0090	0.0120	0.0110	0.0050	0.0040
	swe	-0.0050	0.0140	0.0100	0.0030	0.0070	0.0080	0.0110	0.0100	0.0120	0.0160	0.0170	0.0050	0.0120	0.0210	0.0100
	chi	-0.0450	-0.0730	-0.0430	-0.0400	-0.0670	-0.0230	-0.0330	-0.0600	-0.0560	-0.0740	-0.0630	-0.0540	-0.0400	-0.0490	-0.0500
	dan	-0.0090	-0.0210	0.0210	-0.0160	-0.0200	-0.0030	0.0010	-0.0030	0.0040	-0.0030	-0.0030	0.0100	0.0020	0.0000	0.0020
	eng	-0.0200	0.0050	-0.0250	0.0080	-0.0090	0.0280	0.0130	0.0190	0.0000	-0.0210	-0.0010	-0.0110	0.0070	-0.0060	0.0010
	ger	-0.0240	-0.0020	-0.0100	-0.0490	-0.0360	-0.0590	-0.0250	-0.0300	-0.0470	-0.0230	-0.0240	-0.0430	-0.0470	-0.0340	-0.0360
	heb	0.0280	0.0220	0.0380	0.0560	0.0580	0.0660	0.0580	0.0570	0.0690	0.0650	0.0720	0.0690	0.0700	0.0720	0.0700
	hrv	-0.0120	0.0330	0.0060	0.0290	0.0080	0.0290	0.0160	0.0100	-0.0030	0.0170	0.0200	0.0180	0.0150	0.0210	0.0230
	nor	0.0120	-0.0120	0.0410	0.0270	0.0320	0.0280	0.0380	0.0150	0.0050	0.0170	0.0250	0.0190	0.0160	0.0150	0.0150
	por	0.0080	0.0070	0.0180	0.0100	0.0120	-0.0010	0.0140	0.0180	0.0050	-0.0040	-0.0090	0.0030	-0.0010	0.0040	0.0010
	rom	0.0190	-0.0050	0.0480	0.0530	0.0570	0.0560	0.0650	0.0610	0.0510	0.0490	0.0590	0.0600	0.0550	0.0630	0.0610
	slk	0.0400	0.0410	0.0610	0.0640	0.0700	0.0700	0.0680	0.0660	0.0680	0.0620	0.0660	0.0570	0.0550	0.0630	0.0680
	srb	-0.0140	0.0270	0.0110	0.0200	0.0200	0.0340	0.0320	0.0320	0.0190	0.0330	0.0330	0.0280	0.0230	0.0300	0.0290
	swe	0.0010	0.0120	0.0180	0.0120	0.0210	0.0140	0.0180	0.0120	0.0060	0.0150	0.0030	0.0050	0.0020	0.0050	0.0100

Figure 2: For each target language (outer rows), shows the development of F1-scores for each test language (inner rows) across the 15 training epochs (columns). The numbers and the colors, bounded by -0.2 and 0.2 , of the cells signify the difference $\Delta F1$ relative to the performance for that test language on the baseline model. *Example*: With Chinese as a target language (1st block), the Chinese test set shows consistently better performance (dark green) than the baseline model, while Romanian shows consistently lower span-F1 than the baseline model (dark red), indicating that training with Chinese improved Chinese, while forgetting Romanian.